

Instruction Format

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Instruction Format

An **instruction format** must include an opcode and implicitly or explicitly zero or more operands.

Each explicit operand is referenced using one of the addressing mode that is available for that machine.

The **format** must implicitly or explicitly indicate the addressing mode of each operand.

Introduction to Instructions

Definition: Instruction is a statement by which the operation of CPU is determined.

These instructions referred as “Machine instructions or computer Instructions”

Elements of Machine instruction

- Operation code
- Source operand reference
- Result operand Reference
- Next instruction reference

Elements of Machine instruction

Operation code : Specifies the operation to be performed .The operations is specified by a binary code ,known as the *operation code Or Opcode*.

Source operand reference : The operation may involve one or more source operands that is operands that are input for the operation.

Result operand Reference : The operation may produce a result.

Next instruction reference : This tells the CPU where to fetch the next instruction after the execution of this instruction is complete.

Instruction Representations

Each instruction is represented by sequence of bits

The instruction is divided into two fields

- Opcode field
- Operand field

This operand field further divided into one to four fields.

This layout of the instruction is known as the “Instruction Format”

Simple instruction format

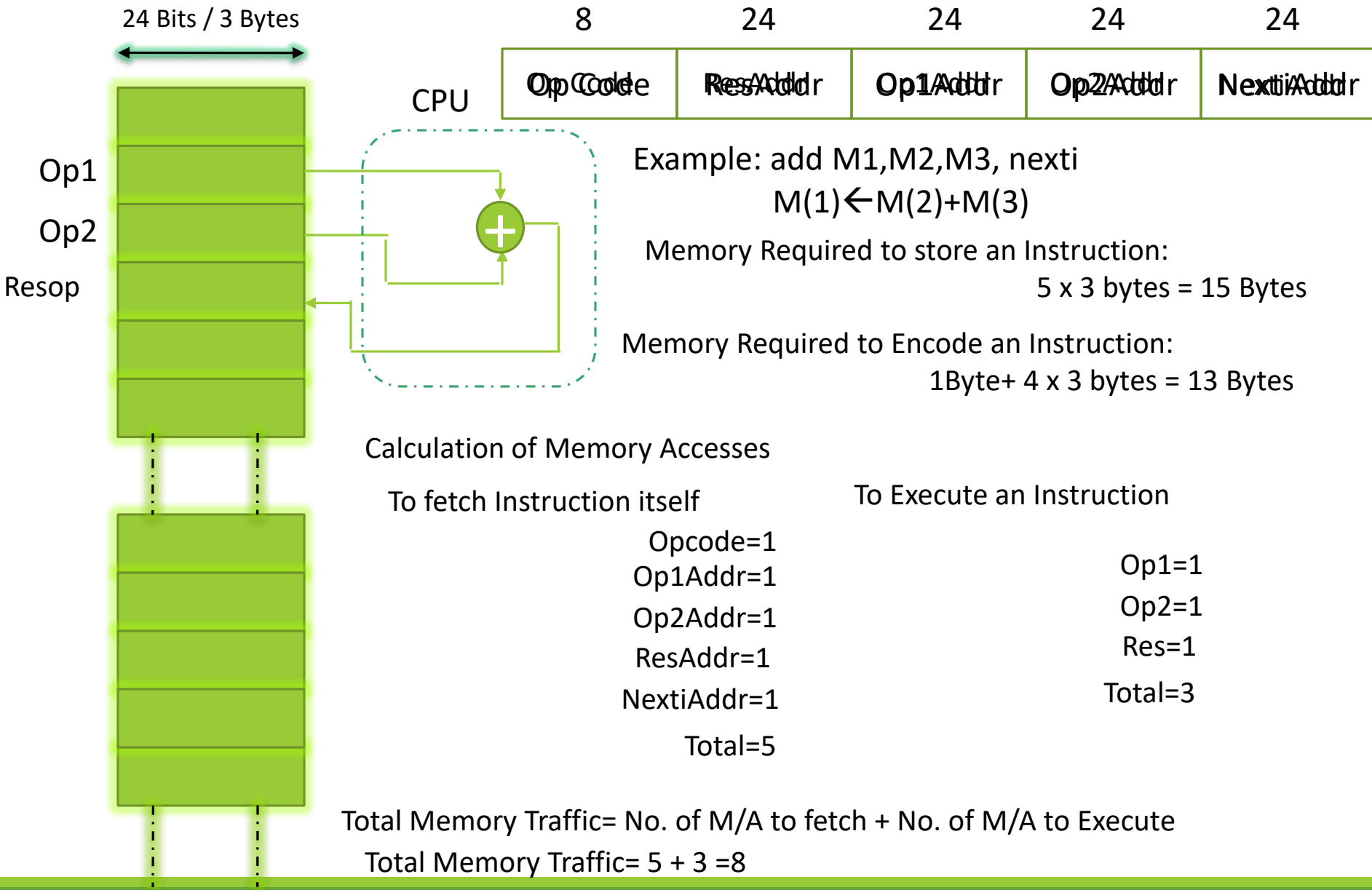
Opcode	Operand Address1	Operand Address2	Result Address1	Next Instruction
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4-,3-,2-,1-,and 0- address instructions

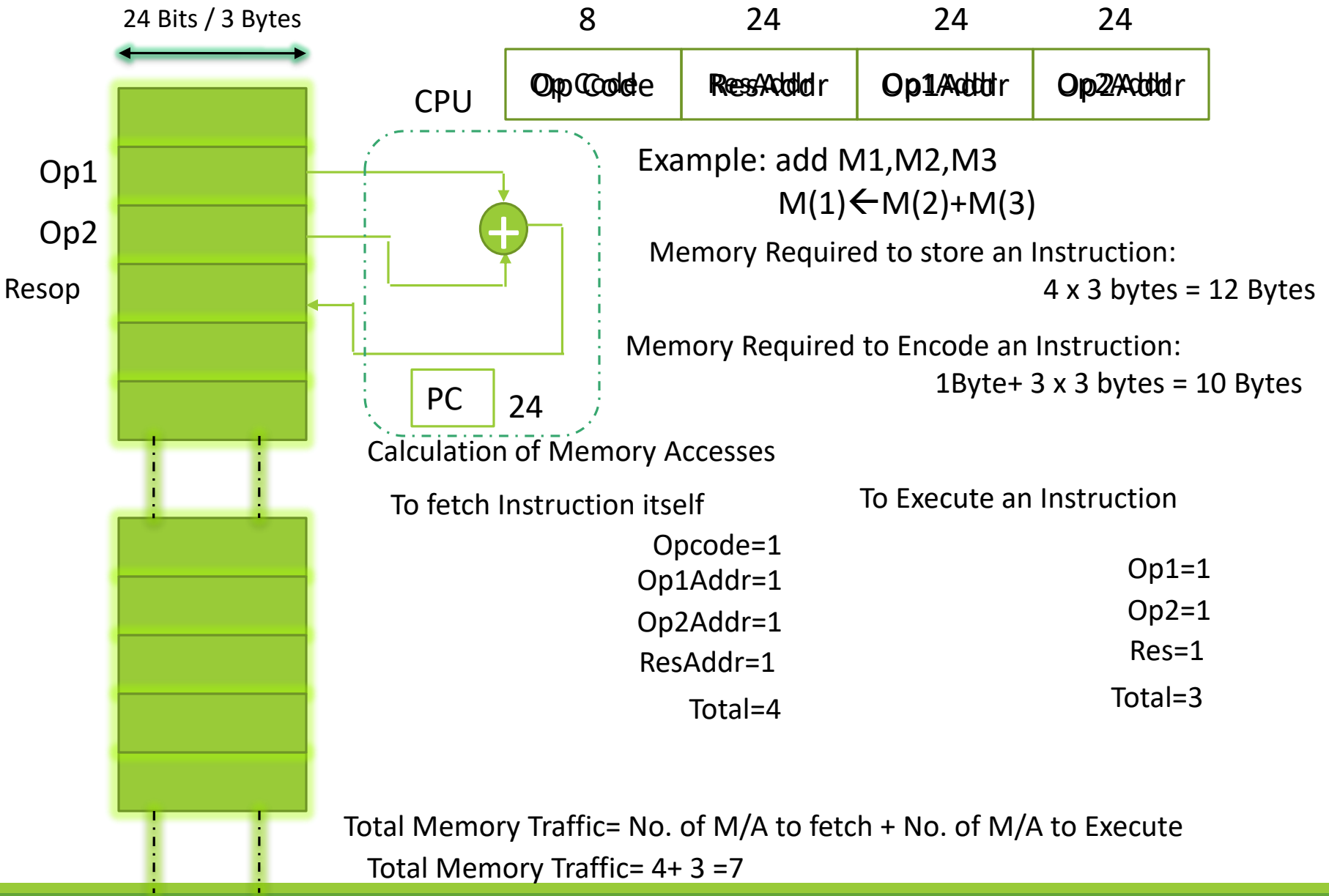
Instruction Set categorized into four categories based on number of operand address in the instruction.

- ☐ 4- Address Instruction
- ☐ 3-Address Instruction
- ☐ 2-Address Instruction
- ☐ 1-Address Instruction
- ☐ 0-Address Instruction

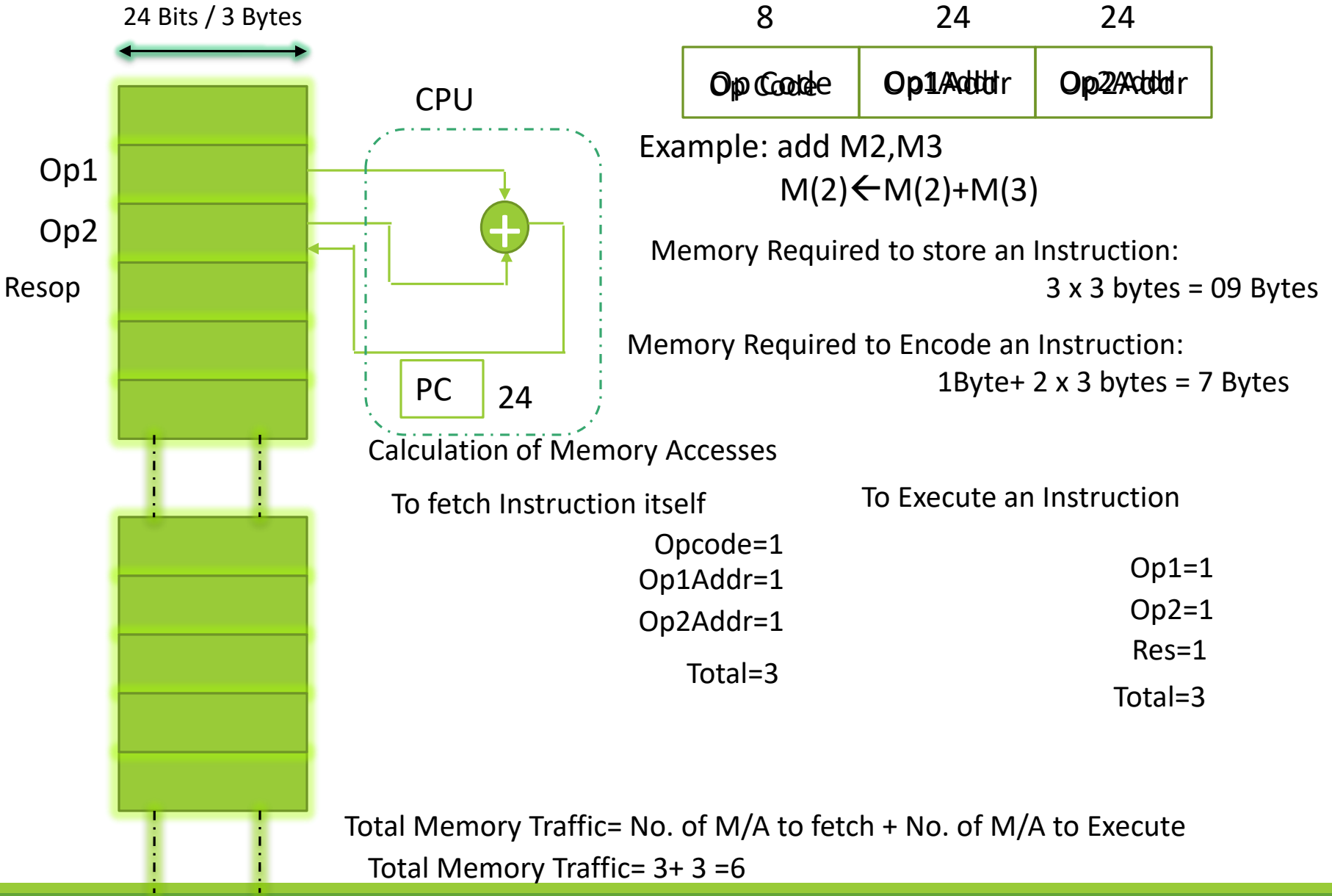
4 Address Instruction



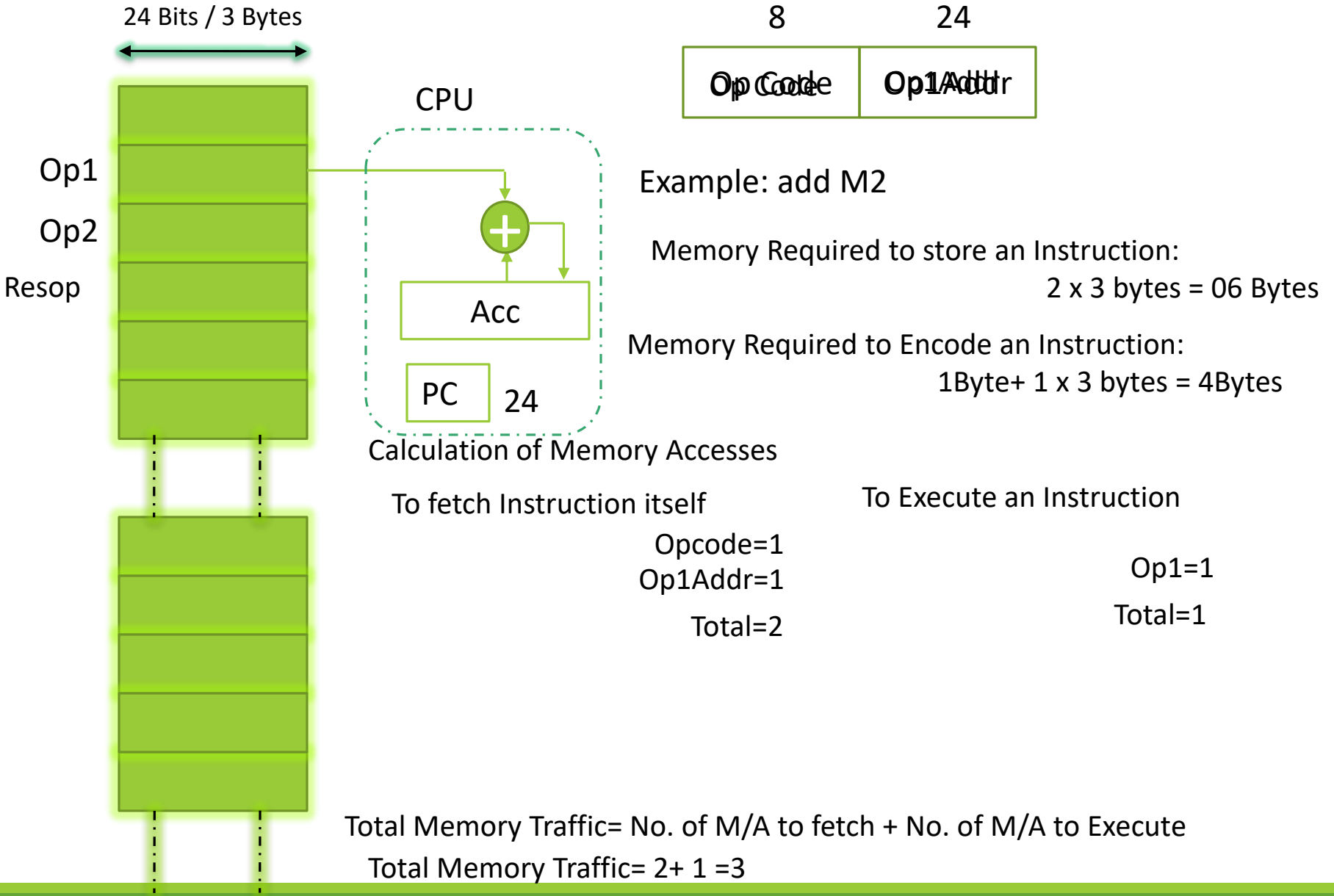
3-Address Instruction



2 Address Instruction

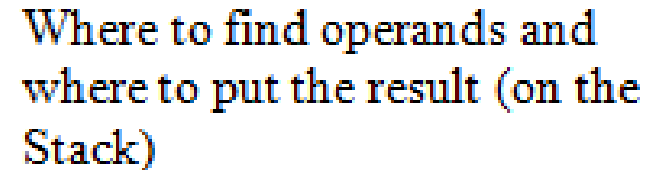


1 Address Instruction



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- Computer must have a 1-address instruction to push and pop operands to and from the stack



# Comparisons

| Instruction Type | Memory To Store in Bytes | Memory To Encode in Bytes | M/As to fetch an Instruction | M/As to Execute an Instruction | Memory Traffic |
|------------------|--------------------------|---------------------------|------------------------------|--------------------------------|----------------|
| 4-address        | $5 \times 3 = 15$        | $1 + (4 \times 3) = 13$   | 5                            | 3                              | $5 + 3 = 8$    |
| 3-Address        | $4 \times 3 = 12$        | $1 + (3 \times 3) = 10$   | 4                            | 3                              | $4 + 3 = 7$    |
| 2-Address        | $3 \times 3 = 09$        | $1 + (2 \times 3) = 07$   | 3                            | 3                              | $3 + 3 = 6$    |
| 1-Address        | $2 \times 3 = 06$        | $1 + (1 \times 3) = 04$   | 2                            | 1                              | $2 + 1 = 3$    |
| 0-Address        | $1 \times 3 = 03$        | $1 + (0 \times 3) = 01$   | 1                            | 0                              | $1 + 0 = 1$    |